Welcome to Junior Riverside Robotics!

We're excited to welcome you to Junior Riverside Robotics—a hands-on club for young creators eager to explore the thrilling world of robotics. Whether you're new to building or already love experimenting with tech, this is your opportunity to collaborate on innovative projects, compete in exciting challenges, and gain practical skills in a supportive environment. As part of the broader Riverside Robotics program, we're geared toward juniors and focused on creativity, teamwork, and real-world problem-solving.

What is Junior Riverside Robotics?

Junior Riverside Robotics is all about designing, building, and battling in the Alliance for Working Together (AWT) RoboBots Competition on April 25, 2026. This season, we're crafting a high-performing 3-pound battle bot ready to dominate the arena—inspired by shows like BattleBots, but powered by your ideas! Our mission is to create a resilient, strategic robot while honing skills in engineering, collaboration, and innovation. Looking ahead, we may expand into broader robotics explorations beyond competitions. We'll share more on rules, strategies, and timelines during our initial meeting. Got questions? Email jack.carbone@riversideschools.net anytime.

This club is proudly sponsored by Show Studio LLC and Arc Technology LLC, with no costs for students—everything from materials to competition is covered.

Meet Your Advisors

We will be guiding you through everything, teaching you everything you are interested in learning pertaining to engineering, design, coding, and more!

Advisors

- **Jack Carbone** Software Engineer
- **Adam Carbone** Electrical Engineer

We cannot wait to get started with this new team!

Skills You'll Build and Apply

Our program integrates key areas of robotics, where mechanical, electrical, software, and documentation skills interconnect to bring your bot to life. No experience required—we'll teach you step by step!

- **Mechanical Engineering**: Use tools like TinkerCAD for CAD modeling to design sturdy frames, weapons, and components. You'll select materials and apply principles to build reliable structures.
- **Electrical Engineering**: Design and test wiring for motors, sensors, batteries, and RC systems. Learn circuit basics, troubleshooting, and safety to keep your bot powered and responsive.
- **Software Engineering**: Code behaviors using block-based or text programming. Develop algorithms to integrate software with hardware for smart, autonomous features.
- **Driving and Strategy**: Hone skills in arena navigation, tactical decision-making, and competitive maneuvering.
- **Pit Crew Expertise**: Master fast repairs and adjustments under time pressure between matches.
- **Documentation**: Keep detailed records of designs, code, schematics, and modifications for team reference, presentations, and future enhancements.

Meeting Schedule

Join us every **Thursday from 5:30 PM to 7:30 PM** in the **Media Center** (sometimes JRW Cafe—stay tuned via the calendar for specifics). Subscribe to our Google Calendar for updates and reminders: Junior Riverside Robotics Calendar.

It is time to gear up for an unforgettable season of innovation and excitement!

Please contact jack.carbone@riversideschools.net with any questions.